SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43
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REG NO:192372098

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: **Total Marks: 30**

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.
5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

*I **G.CHAKRADHAR(192311439)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

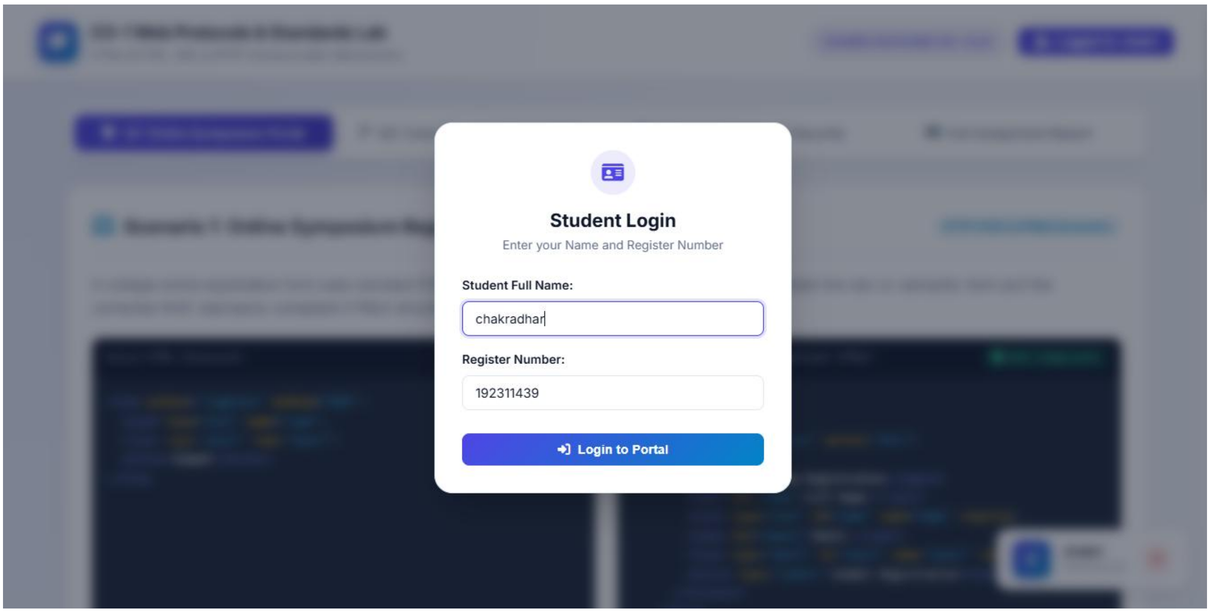
Signature of the student: E.HEMANTH REDDY

RUBRICS FOR CALCULATION:

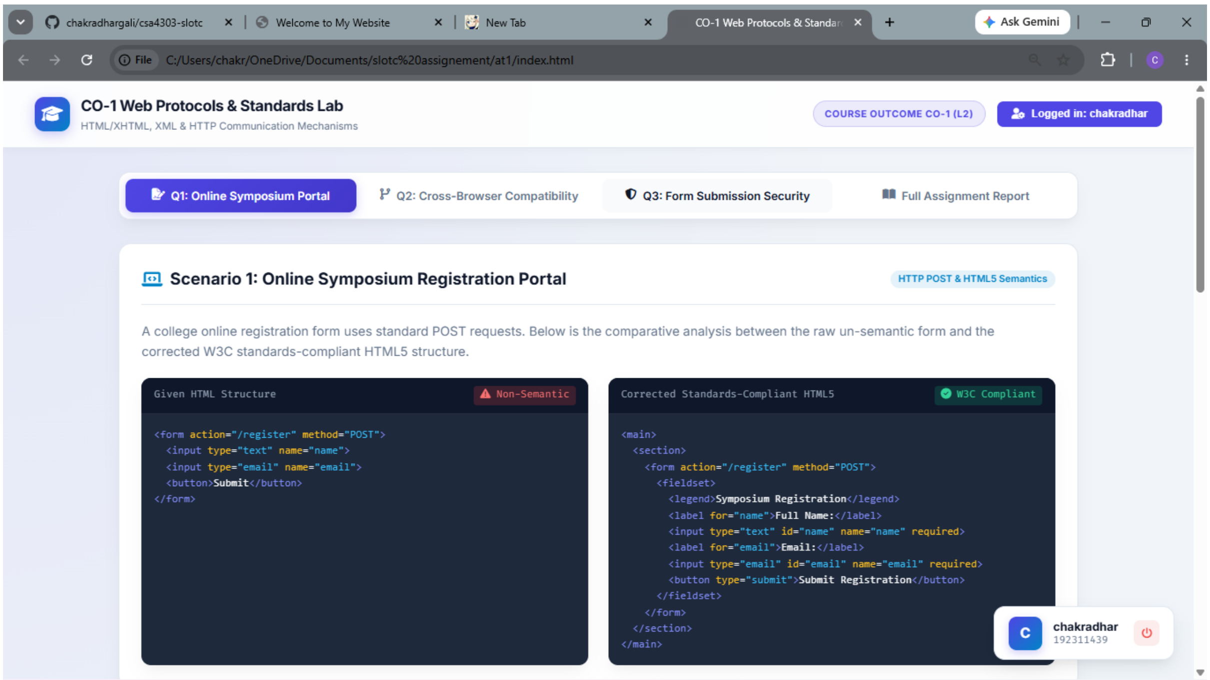
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

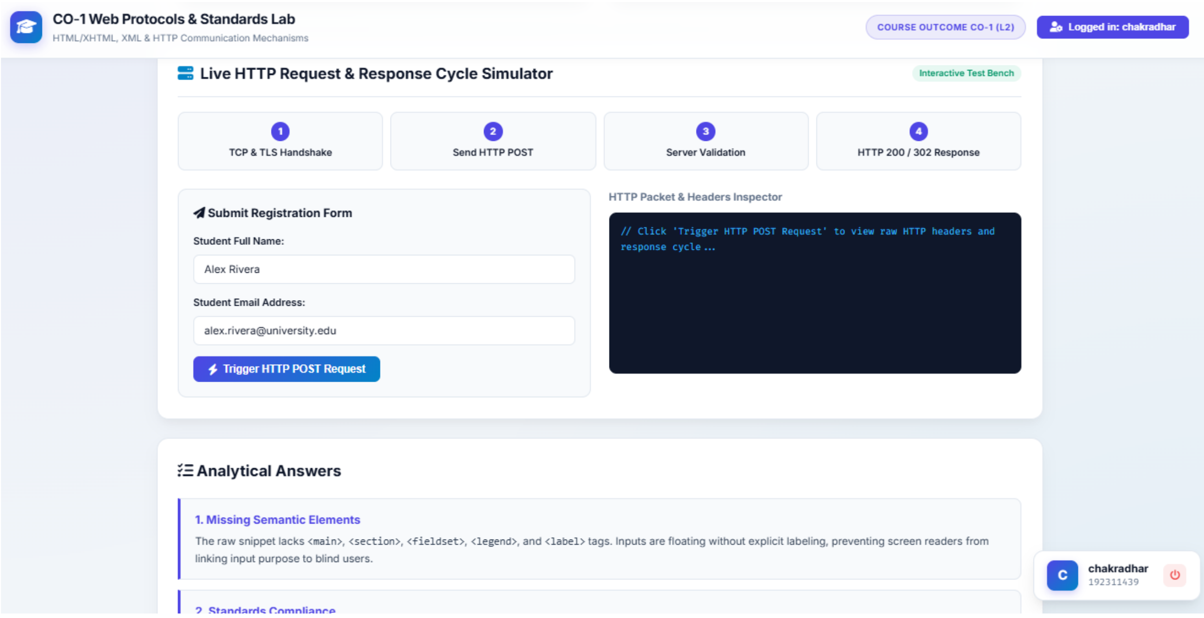
OUTPUT:

1.login:

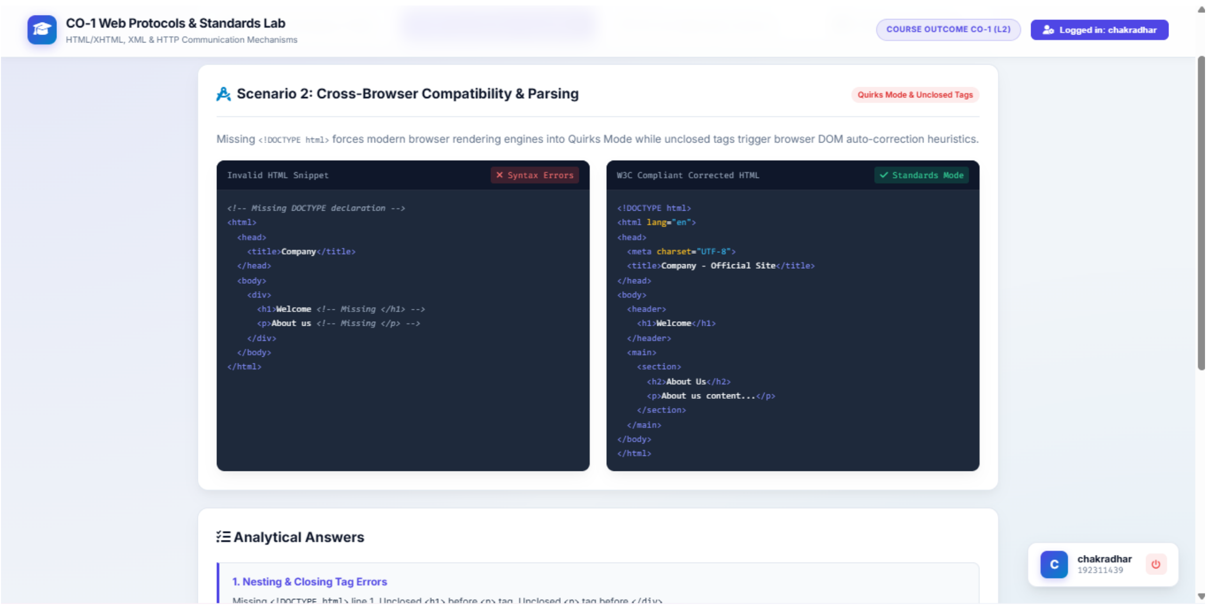


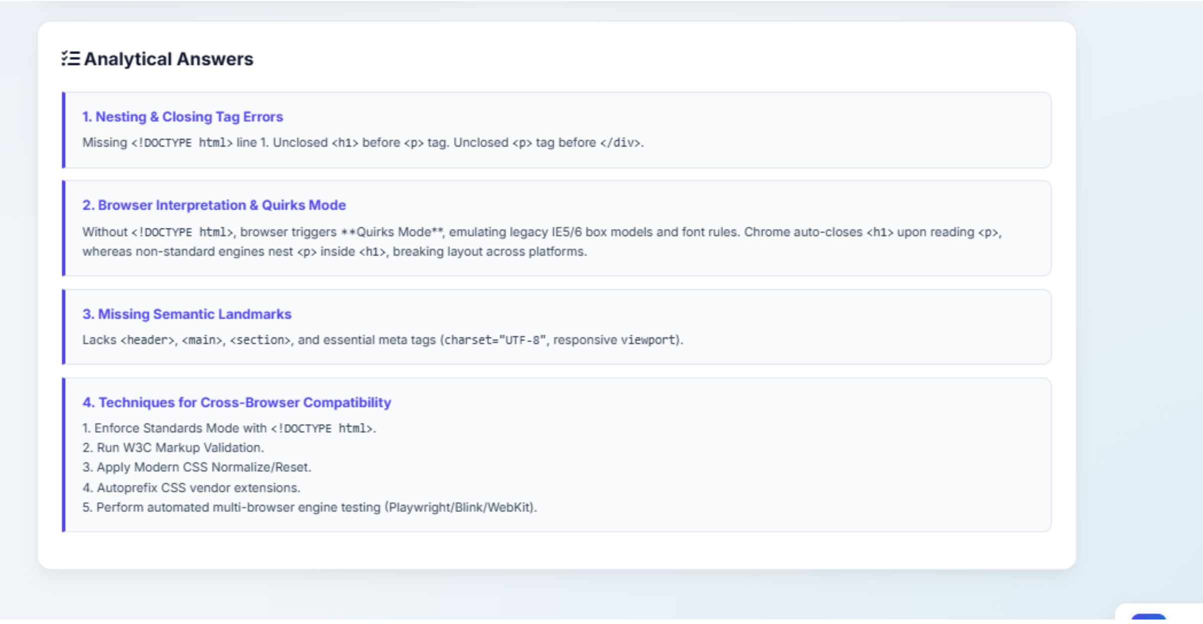
2.question-1:




CO-1 Web Protocols & Standards Lab
HTML/XHTML, XML & HTTP Communication Mechanisms
COURSE OUTCOME CO-1 (L2)
Logged in: chakradhar
Live HTTP Request & Response Cycle Simulator
Interactive Test Bench
1
TCP & TLS Handshake
2
Send HTTP POST
3
Server Validation
4
HTTP 200 / 302 Response
Submit Registration Form
Student Full Name:
Alex Rivera
Student Email Address:
alex.rivera@university.edu
Trigger HTTP POST Request
HTTP Packet & Headers Inspector
// Click 'Trigger HTTP POST Request' to view raw HTTP headers and response cycle...
Analytical Answers
1. Missing Semantic Elements
The raw snippet lacks <main>, <section>, <fieldset>, <legend>, and <label> tags. Inputs are floating without explicit labeling, preventing screen readers from linking input purpose to blind users.
2. Standards Compliance
chakradhar
192311439

Question2:


CO-1 Web Protocols & Standards Lab
HTML/XHTML, XML & HTTP Communication Mechanisms
COURSE OUTCOME CO-1 (L2)
Logged in: chakradhar
Scenario 2: Cross-Browser Compatibility & Parsing
Quirks Mode & Unclosed Tags
Missing <!DOCTYPE html> forces modern browser rendering engines into Quirks Mode while unclosed tags trigger browser DOM auto-correction heuristics.
Invalid HTML Snippet
✕ Syntax Errors
<!-- Missing DOCTYPE declaration -->
<html>
<head>
<title>Company</title>
</head>
<body>
<div>
<h1>Welcome <!-- Missing </h1> -->
<p>About us <!-- Missing </p> -->
</div>
</body>
</html>
W3C Compliant Corrected HTML
✓ Standards Mode
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Company - Official Site</title>
</head>
<body>
<header>
<h1>Welcome</h1>
</header>
<main>
<section>
<h2>About Us</h2>
<p>About us content...</p>
</section>
</main>
</body>
</html>
Analytical Answers
1. Nesting & Closing Tag Errors
chakradhar
192311439



Analytical Answers

1. Nesting & Closing Tag Errors

Missing `<!DOCTYPE html>` line 1. Unclosed `<h1>` before `<p>` tag. Unclosed `<p>` tag before `</div>`.

2. Browser Interpretation & Quirks Mode

Without `<!DOCTYPE html>`, browser triggers **Quirks Mode**, emulating legacy IE5/6 box models and font rules. Chrome auto-closes `<h1>` upon reading `<p>`, whereas non-standard engines nest `<p>` inside `<h1>`, breaking layout across platforms.

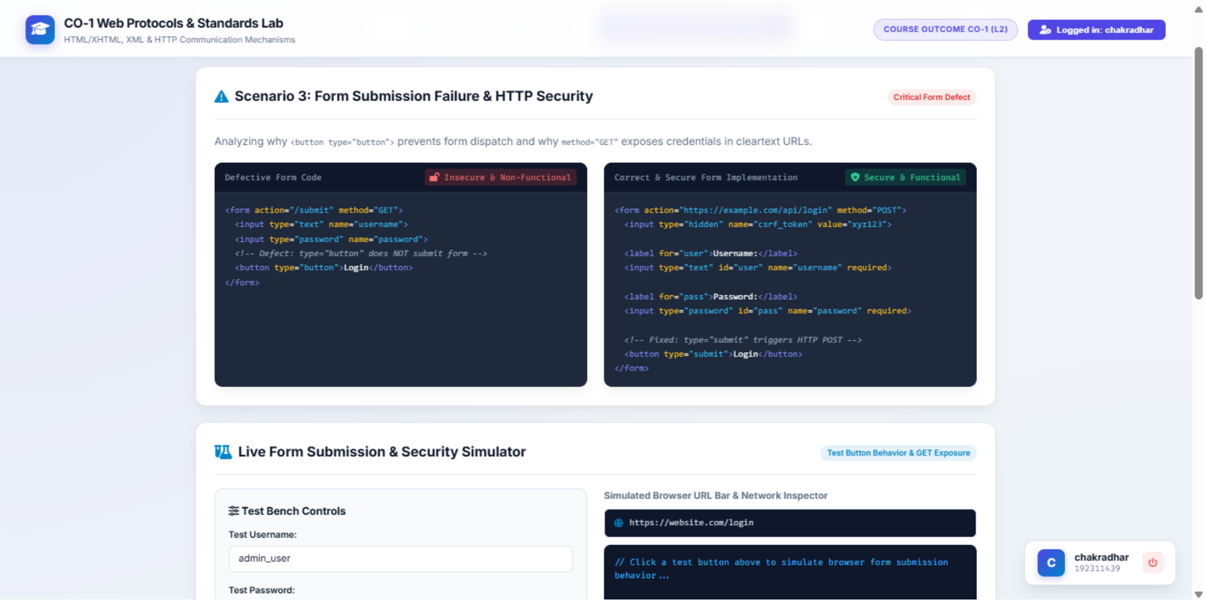
3. Missing Semantic Landmarks

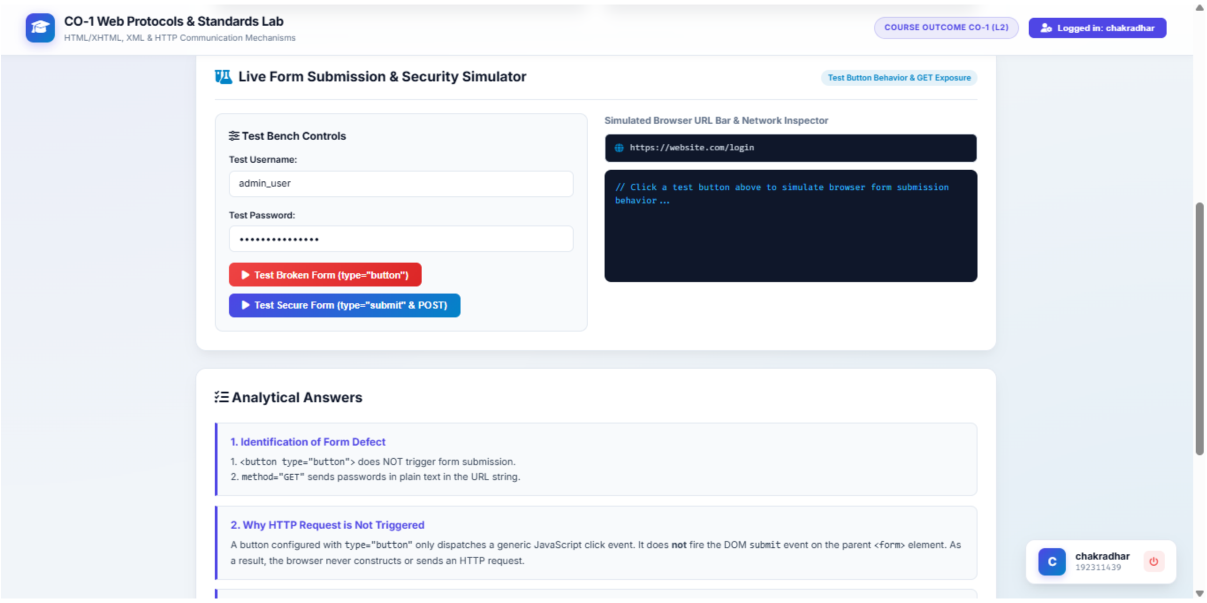
Lacks `<header>`, `<main>`, `<section>`, and essential meta tags (`charset="UTF-8"`, responsive `viewport`).

4. Techniques for Cross-Browser Compatibility

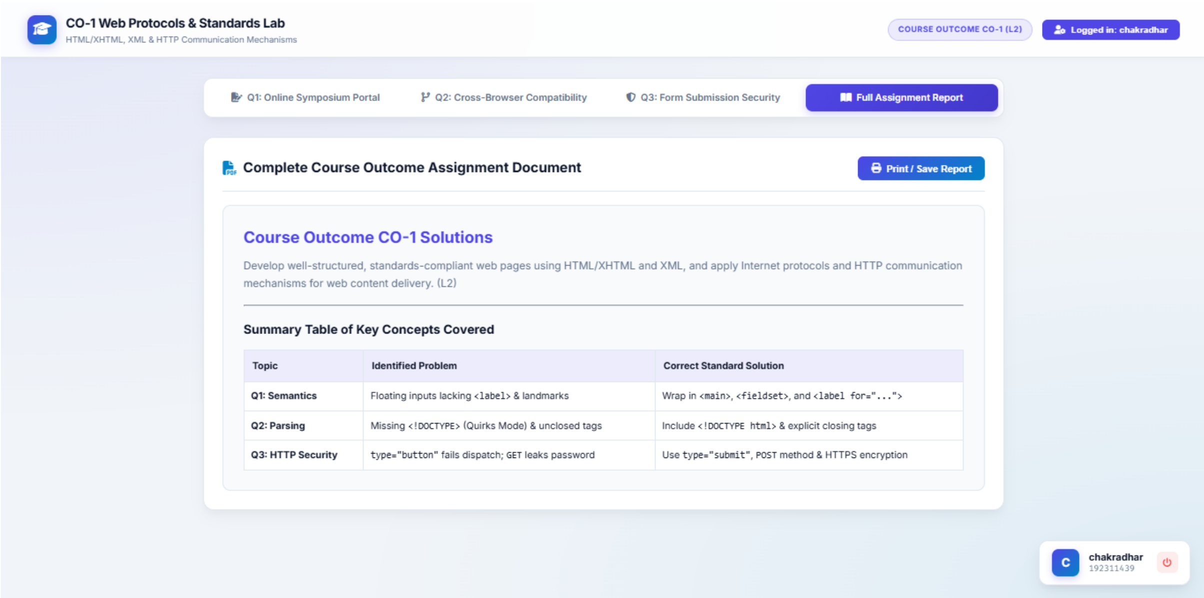
1. Enforce Standards Mode with `<!DOCTYPE html>`.
2. Run W3C Markup Validation.
3. Apply Modern CSS Normalize/Reset.
4. Autoprefix CSS vendor extensions.
5. Perform automated multi-browser engine testing (Playwright/Blink/WebKit).

Question 3:





Full assignment report:



Result:

We have successfully Developed well-structured, standards-compliant web pages using HTML/XHTML and XML, and we applied Internet protocols and HTTP communication mechanisms for web content delivery